

everything matches. Once this is verified, this shared data is then discarded. What they are left with is a shorter string of data that only the two of them know—the Key!

But wait, didn't we say that Alice and Bob discuss all this in *public*? What happened about all that secrecy that was promised? Let's take a look at what information Eve has if she was listening in on all of this (and of course she was):

1. She knows which machines both Alice and Bob used, and

2. She knows a few selected bits of the key.

Our response:

1. So what if she knows which machines were used? She doesn't know what answers the machines gave Alice and Bob, does she?

2. Again, so what? Remember, the bits that are discussed in public are discarded and are no longer part of the key.

Take That, Eve!

Our concern here is a man-in-the-middle attack (see box *Attack!*). How can Eve find out the key without both Alice and Bob knowing she's there? She needs to send them her own cards, and make sure that both Alice and Bob get the result that Eve wants them to. Suppose she sends out a pair of cards with 1 on them, and both Alice and Bob use the blue machine (Eve can find this out because Alice and Bob are going to discuss what machines they used). So because both Alice and Bob used the blue machine, they got 1 as their output—exactly what Eve wanted. If this continued this way, Alice and Bob would end up agreeing on a key that Eve created. Bye bye, privacy!

Isn't there a possibility that Alice and Bob could end up using different machines? Their output wouldn't tally, and Eve can't know what the outcome will be. True, but these results are discarded anyway—they didn't use the same machine! So all Eve needs to do is skip this part in the key she's trying to send Alice and Bob, and see what they got for the next bit. She'd still know the part of the key that *has* been agreed upon, so she's still a threat. Could it be that a quantum cryptography system is not immune to attacks?

Wait! What if Eve sent out blue cards, and Alice and Bob used their *red* machines? Would they get the same result then? Now this is where Eve's dirty little secret is revealed—you see, she didn't send out entangled cards—just identical cards to fool Alice and Bob into believing they are entangled. So there's a 50 per cent chance that even though Alice and Bob use the same machine, they will get different answers. When they do their final check and reveal parts of the key to each other, they are going to end up in a situation where they both used red, but got different answers. "Something's wrong!" they'd say, "let's shut up for now." And that's the end of Eve's game.

Reality Check

The concept of quantum cryptography has been around since the late 1970s, and working systems have been set up in educational institutions as early as the 90s. However, it will take

a while before we see it in more places. For one thing, quantum cryptography is possible only over fiber-optic lines. Before we see it in our daily lives, we still need to see a world where fiber-optics has penetrated to a much greater degree. Entanglement is a fun concept to read about, but it is a *really* difficult thing to do, and even once it's done, controlling that entanglement is possible only over small distances—around 60 km or so.

Most importantly, quantum cryptography hasn't evolved to suit a network scenario—point-to-point connections are all there is. Even so, there are real-life applications where it is already starting to become practical.

My Preciousssss...

The infrastructure for a quantum cryptography system is a massive expense; whose information could possibly be *so* secret that they wouldn't mind shelling out obscene amounts of money to protect it? Why, the military, of course! We don't want our national secrets being bandied about on public forums, do we?

Think: ten years from now, you're making a withdrawal from your ATM. You need to punch in your PIN, don't you? Would you really be comfortable with the idea that there could be a hacker tapping the line, waiting to make his quantum computer decipher your PIN?

Companies such as New York's MagiQ and Geneva's idQuantique are already providing quantum cryptography solutions to banks and the military—obviously the first people who would want such ironclad protection. Research is underway to use quantum cryptography in communication with satellites as well.

Billions of dollars are spent in credit card transactions on the Internet today. Sure, the cryptography used is really secure—the time it would take to decrypt your credit card number is much, much longer than the actual validity of your card. For example, it is estimated that it would take around a thousand years to break current techniques using today's computers. Who cares, then, if you're going to be long gone before your credit card number is found out? But then again, there is the looming threat of quantum computers. As we move on, and quantum cryptography can be used on the Internet, we should see an almost total stop to things like credit-card theft; almost every financial transaction could be carried out over the Net without any worries.

Another Reality Check

Quantum cryptography may be the ultimate safeguard against eavesdroppers—even eavesdroppers with quantum computers. There is still the matter of encryption on your hard drive—if a hacker with a quantum computer gets into your system, nothing says that he can't break any kind of file encryption you have used. Secrets—especially military secrets—need to be kept hidden for decades, and we still need a system for encryption in storage media before everything is truly under lock and key. ☒

nimish_chandiramani@thinkdigit.com